ASMI BASNET

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PROFESSIONAL SUMMARY

AI Developer and Software Systems Engineer (M.S. in Analytics candidate, Saint Louis University; B.E. in Electronics & Communication) with 3+ years of experience designing, building, and deploying production AI solutions, generative AI applications, and resilient backend microservices. Proficient in Python, C/C++, TypeScript, and SQL, with hands-on expertise engineering retrieval-augmented generation (RAG) pipelines, tool-calling multi-agent systems (Agno, WebMCP), and RESTful APIs (FastAPI). Proven track record implementing automated model evaluation frameworks, citation verification guardrails, and data quality assurance suites (IV&V) that eliminate hallucinations and detect schema drift. Adept at collaborating in Agile sprints across cross-functional engineering, data, and product teams to translate complex requirements into maintainable, secure software adhering to responsible AI governance.

TECHNICAL SKILLS & CORE COMPETENCIES

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), OpenAI GPT-4o API, Anthropic Claude API (Prompt Caching & Tools), Local LLM Inference (Ollama: Qwen 2.5, Llama 3, Phi-3), Agno Multi-Agent Framework, Model Context Protocol (WebMCP), Vector Databases (ChromaDB, FAISS), Citation Verification, Hallucination Guardrails, Context Optimization.

Programming & Software: Python (FastAPI, Flask, Requests, Pandas, NumPy, Scikit-Learn, PyTorch, SQLAlchemy), JavaScript / TypeScript, C/C++, Go (Golang), SQL (PostgreSQL, DuckDB, Snowflake, SQL Server: CTEs, Window Functions), RESTful API Design, Webhooks, Linux CLI, Git / GitHub, Docker, Docker Compose, CI/CD Actions.

Machine Learning & Evaluation: Supervised & Unsupervised ML (Scikit-Learn, XGBoost, LightGBM, Random Forest, Lasso), Deep Learning (PyTorch CNNs, LSTM), Explainable AI (SHAP TreeExplainer), Time-Series Forecasting (Prophet, ARIMA), K-Fold Cross-Validation, Model Evaluation (ROC-AUC, Precision, Recall, F1).

Data Governance & Quality: Automated Data Quality Assurance (IV&V), Schema Drift Monitoring, Data Profiling, Audit Logging, Responsible AI & Data Privacy (Air-Gapped / Zero-Leakage Architectures), Star-Schema Dimensional Modeling, Business Intelligence (Microsoft Power BI, DAX, Tableau).

PROFESSIONAL EXPERIENCE

Saint Louis University — **St. Louis, MO**

Graduate AI Developer & Research Assistant (SPS-Corps) — *August 2025 – Present*

- Architected and deployed an air-gapped, offline RAG compliance application in Python (FastAPI, ChromaDB, Ollama) that runs local quantized LLMs to analyze confidential enterprise documentation with zero third-party API exposure and zero data leakage.
- Engineered a chain-of-thought verification engine evaluating retrieved context sentence-by-sentence, flagging contradictory requirements (FLAG_CONFLICT), and requesting missing context (CLARIFY) to automate governance audits.
- Designed cooperating multi-agent automation workflows leveraging the Agno framework and Anthropic Claude / OpenAI APIs, integrating custom tool-calling routines and REST API connectors that automated cross-departmental research and data synthesis, cutting manual compilation cycles by 90%.
- Engineered automated data ingestion, validation, and ETL pipelines across 8+ operational databases using Python and modular SQL CTEs with window functions, standardizing data dictionaries and cutting query execution times by 40%.
- Built a hybrid forecasting and anomaly detection engine pairing Facebook Prophet with PyTorch LSTM neural networks, automatically flagging demand surges and budget variances beyond ±2.5 standard deviations to deliver predictive resource allocation insights.
- Partnered with engineering and administrative leadership in Agile sprints to translate operational requirements into technical specifications, authoring runbooks and conducting hands-on user training workshops for 25+ staff members to drive seamless tool adoption.

Saint Louis University — **St. Louis, MO**

Data & AI Solutions Analyst Intern — *June 2024 – August 2024*

- Built and deployed 6 interactive executive dashboards across Power BI and Tableau backed by star-schema data models, consolidating 15+ operational KPIs to enable leadership to track resource utilization, productivity, and project milestones.
- Engineered automated data profiling, cleaning, and validation scripts in Python (Pandas) and SQL across 10,000+ institutional records, establishing automated webhook alerts that instantly flag schema drift and transactional anomalies.
- Enforced strict institutional data governance and privacy rules, building row-level data quality validation checks across reporting layers.

Entegra Sources Pvt. Ltd. (Mayahold Group) — **Remote / Kathmandu, Nepal**

Systems & Data Analyst — *December 2022 – July 2023*

- Extracted, cleaned, and normalized multi-source operational and telemetry datasets (5,000+ records) into standardized 3NF relational schemas using Python (Pandas, SQLAlchemy) and SQL, ensuring complete data consistency across enterprise systems.
- Validated algorithmic logic and automated backend processing scripts in Python and C/C++, implementing robust exception handling, boundary checks, and audit logging to maintain 99.8% pipeline reliability.
- Authored formal technical documentation detailing data dictionaries, data validation rules, and sampling methodologies to establish organizational data governance standards.

FEATURED AI ENGINEERING & AUTOMATION PROJECTS

PolicyGuard — Enterprise Offline RAG & Document Intelligence Suite — **Python, Ollama, ChromaDB, FastAPI, Docker**

- Engineered an air-gapped, privacy-first RAG application running entirely on local quantized LLMs via Ollama to analyze confidential enterprise policy documents and contracts with zero data leakage.
- Implemented automated chain-of-thought citation verification and dynamic cosine similarity guardrails, evaluating retrieved context sentence-by-sentence to detect contradictory clauses (FLAG_CONFLICT) and eliminate manual review overhead.

Nepal Disaster Watch — WebMCP Real-Time Autonomous Command Center — **TypeScript, WebMCP, REST APIs, GeoJSON**

- Architected an AI-driven crisis intelligence command center featuring 11 native WebMCP tools (6 read, 5 write) for OpenAI's WebMCP Challenge, synchronizing 6 live institutional feeds across all 77 districts of Nepal.
- Engineered resilient feed status adapters (live, snapshot, unreachable), allowing autonomous AI agents to dynamically query spatial risks, pan map layers, and prevent silent system failures during critical operations.

Autonomous Multi-Agent Workflow & Research System — **Python, Agno Framework, OpenAI GPT-4o / Claude API, Zapier**

- Built a cooperating multi-agent workflow pairing a Market Scout Agent with an Analytical Synthesis Agent to autonomously monitor data feeds, call external APIs, evaluate valuation anomalies, and synthesize structured executive briefs.
- Integrated event-driven Zapier webhooks and automated messaging endpoints to ingest real-time operational triggers and distribute structured summaries, reducing manual information gathering and reporting cycles by 90%.

EDUCATION & CERTIFICATIONS

Saint Louis University | Master of Science in Analytics (STEM) — **St. Louis, MO | Expected May 2026**

Tribhuvan University | Bachelor of Engineering in Electronics & Communication (STEM) — **Kathmandu, Nepal | Conferred 2023**

Certifications: Building with Claude & Anthropic API (Prompt Caching & Tools) · AI Engineer for Data Scientists Associate (DataCamp: PyTorch & Scikit-Learn) · Microsoft Certified: Power BI Data Analyst Associate (PL-300) · Project Management & Agile Certification (GlobalShala & SLU).